

DIRECTORY

The earliest known scientists appeared in the ancient world, their discoveries immortalized by the advent of writing. Pioneering individuals from flourishing civilizations started devising methods to observe and control nature, cultivating disciplines such as medicine and astronomy in the process.

DEMOCRITUS

c.460–370 BCE

Greek philosopher Democritus, along with his teacher Leucippus, proposed the idea that everything is made up of tiny, unchanging particles called “atoms,” which move around freely and combine with other atoms to form different objects. This theory, known as atomism, was the first view of the Universe that did not rely on the notion of a god or gods. Democritus was a modest man who worked hard and acquired fame not for his philosophy, but for his ability to predict changes in the weather. He was known as the “laughing philosopher” for his tendency to laugh at the foolishness of others.

EUCLID

325–c.265 BCE

Often called the “father of geometry,” Greek mathematician Euclid is best known for his *Elements of Geometry*, a compilation of the advances made in Greek mathematics over the previous 300 years. *Elements* was translated into Arabic in about 800 CE, and in 1482, it became the first mathematics textbook to be printed. Euclid’s principles formed the basis of mathematical teaching in

the West and Arab world for more than 2,000 years. Although little is known about his life, Euclid is also said to have written about optics and astronomy.

HIPPARCHUS

190–c.120 BCE

When he was compiling his catalog of stars in 129 BCE, Greek astronomer Hipparchus noticed that the positions of the stars he saw were different from the earlier records he was consulting. He concluded that it was not the stars that had moved, but Earth itself, and so proposed axial precession—a gradual but continuous change in the orientation of Earth’s rotational axis. Hipparchus is also said to have invented trigonometry—the branch of mathematics that deals with calculating the angles of triangles and the lengths of their sides.

PLINY THE ELDER

23–79 CE

Pliny the Elder was born in Como, in what is now Italy, and served as a military commander for the Roman Empire. He spent much of his spare time writing and studying, but his only surviving work is *Natural History*, which

he completed in 77 CE. Made up of 37 books, *Natural History* covered an unprecedented breadth of ancient knowledge, including botany, astronomy, geography, zoology, and medicine, and became a model for later encyclopedias. Based in the Bay of Naples on his last military assignment, Pliny died after inhaling fumes from the eruption of Mount Vesuvius.

LIU HUI

c.225–298 CE

Little is known about the Chinese mathematician Liu Hui except that he lived in the northern Wei kingdom and wrote two works. The most famous of these was his commentary on *The Nine Chapters on the Mathematical Art*, an early Chinese mathematics handbook that offered mathematical solutions to everyday problems. Liu showed that the methods suggested in the book were correct, and in doing so, he created proofs that validated mathematical methods for establishing the area of circles and the volume of solids such as pyramids, as well as for adding fractions.

BRAHMAGUPTA

597–668 CE

Indian astronomer and mathematician, Brahmagupta grew up in Bhillamala, in northwest India, and went on to become head of the astronomical observatory at Ujjain, in central India. His two major works on astronomy and mathematics were written entirely in